In[∘]:=
```
(*PRÁCTICA OSCILACIONES ACOPLADAS*)

(*3*)
k = 10.15003805;
```

In[∘]:=
```
ek = 0.000339653;
m = 0.251663333;
em = 0.001607745;
```

In[∘]:=
```
(*Simétrico*)
D[Sqrt[(1 - 1 / Sqrt[3]) * 3 * k / (2 * m)], k]
```
raíz cuadrada raíz cuadrada

Out[∘]=

$$\frac{\sqrt{\frac{3}{2}\left(1-\frac{1}{\sqrt{3}}\right)}}{2\sqrt{\frac{k}{m}}\ m}$$

In[∘]:=
```
D[Sqrt[(1 - 1 / Sqrt[3]) * 3 * k / (2 * m)], m]
```
raíz cuadrada raíz cuadrada

Out[∘]=

$$-\frac{\sqrt{\frac{3}{2}\left(1-\frac{1}{\sqrt{3}}\right)}\ k}{2\sqrt{\frac{k}{m}}\ m^2}$$

In[∘]:=

$$\text{errw1} = \text{Sqrt}\left[\left(\frac{\sqrt{\frac{3}{2}\left(1-\frac{1}{\sqrt{3}}\right)}}{2\sqrt{\frac{k}{m}}\ m} * \text{ek}\right)\text{^}2 + \left(-\frac{\sqrt{\frac{3}{2}\left(1-\frac{1}{\sqrt{3}}\right)}\ k}{2\sqrt{\frac{k}{m}}\ m^2} * \text{em}\right)\text{^}2\right]$$

raíz cuadrada

Out[∘]=

0.0161522

```
(*Antisimétrico*)
```

In[∘]:=
```
D[Sqrt[(1 + 1 / Sqrt[3]) * 3 * k / (2 * m)], k]
```
raíz cuadrada raíz cuadrada

Out[∘]=

$$\frac{\sqrt{\frac{3}{2}\left(1+\frac{1}{\sqrt{3}}\right)}}{2\sqrt{\frac{k}{m}}\ m}$$

In[∘]:=
```
D[Sqrt[(1 + 1 / Sqrt[3]) * 3 * k / (2 * m)], m]
```
raíz cuadrada raíz cuadrada

Out[∘]=

$$-\frac{\sqrt{\frac{3}{2}\left(1+\frac{1}{\sqrt{3}}\right)}\ k}{2\sqrt{\frac{k}{m}}\ m^2}$$

In[ ]:=
```
errw2 = Sqrt[(Sqrt[3/2 (1 + 1/Sqrt[3])]/(2 Sqrt[k/m] m) * ek)^2 + (-(Sqrt[3/2 (1 + 1/Sqrt[3])] k)/(2 Sqrt[k/m] m^2) * em)^2]
```
raíz cuadrada

$$\text{errw2} = \text{Sqrt}\left[\left(\frac{\sqrt{\frac{3}{2}\left(1+\frac{1}{\sqrt{3}}\right)}}{2\sqrt{\frac{k}{m}}\,m} * ek\right)^2 + \left(-\frac{\sqrt{\frac{3}{2}\left(1+\frac{1}{\sqrt{3}}\right)}\,k}{2\sqrt{\frac{k}{m}}\,m^2} * em\right)^2\right]$$

In[ ]:=
```
0.03120374718348187`
m
```

Out[ ]= 0.0312037

Out[ ]= 0.251663

In[ ]:=
```
Clear[k, ek, m, em, errw1, errw2]
```
borra

In[ ]:=
```
(*4*)
m = 0.2501;
k2 = 114.5087753;
k = 10.15003805;
ek = 0.000339653;
em = 0.00065;
ek2 = 0.009971674;
(*Simétrico*)
D[Sqrt[k / m], k]
```
·· raíz cuadrada

General: 10.15003805` is not a valid variable.

Out[ ]= $\partial_{10.15}6.37055$

In[ ]:=
```
D[Sqrt[k / m], m]
```
·· raíz cuadrada

Out[ ]=

$$-\frac{k}{2\sqrt{\frac{k}{m}}\,m^2}$$

In[ ]:=
```
errw1 = Sqrt[(1/(2 Sqrt[k/m] m) * ek)^2 + (k/(2 Sqrt[k/m] m^2) * em)^2]
```
raíz cuadrada

$$\text{errw1} = \text{Sqrt}\left[\left(\frac{1}{2\sqrt{\frac{k}{m}}\,m} * ek\right)^2 + \left(\frac{k}{2\sqrt{\frac{k}{m}}\,m^2} * em\right)^2\right]$$

Out[ ]= 0.00827909

In[ ]:=
```
(*Antisimetrico*)
D[Sqrt[(k + 2 * k2) / m], k]
```
·· raíz cuadrada

Out[ ]=

$$\frac{1}{2\sqrt{\frac{k+2\,k2}{m}}\,m}$$

In[ ]:= D[Sqrt[(k + 2 k2) / m], m]
raíz cuadrada

Out[ ]=
$$-\frac{k + 2\,k2}{2\,\sqrt{\frac{k+2\,k2}{m}}\,m^2}$$

In[ ]:= D[Sqrt[(k + 2 k2) / m], k2]
raíz cuadrada

Out[ ]=
$$\frac{1}{\sqrt{\frac{k+2\,k2}{m}}\,m}$$

In[ ]:= errw2 = Sqrt$\left[\left(\frac{1}{2\sqrt{\frac{k+2\,k2}{m}}\,m} * ek\right)^{\wedge}2 + \left(-\frac{k+2\,k2}{2\sqrt{\frac{k+2\,k2}{m}}\,m^2} * em\right)^{\wedge}2 + \left(\frac{1}{\sqrt{\frac{k+2\,k2}{m}}\,m} * ek2\right)^{\wedge}2\right]$
raíz cuadrada

Out[ ]=
0.0402057

In[ ]:= Clear[k, k2, m, em, ek, errw1, errw2]
borra

```
In[ ]:= (*6*)
(*w1*)
k = 10.15003805;
m =  0.24948;
em = 0.000630159;
ek = 0.000339653;
```

In[ ]:= D[Sqrt[2 * k / m], k]
raíz cuadrada

Out[ ]=
$$\frac{1}{\sqrt{2}\,\sqrt{\frac{k}{m}}\,m}$$

In[ ]:= D[Sqrt[2 * k / m], m]
raíz cuadrada

Out[ ]=
$$-\frac{k}{\sqrt{2}\,\sqrt{\frac{k}{m}}\,m^2}$$

In[ ]:= errw1 = Sqrt$\left[\left(\frac{1}{\sqrt{2}\sqrt{\frac{k}{m}}\,m} * ek\right)^{\wedge}2 + \left(-\frac{k}{\sqrt{2}\sqrt{\frac{k}{m}}\,m^2} * em\right)^{\wedge}2\right]$
raíz cuadrada

Out[ ]=
0.0113934

In[∘]:= (*w2*)
D[Sqrt[(2 - Sqrt[2]) * k / m], k]
⌊·· ⌊raíz cuad··· ⌊raíz cuadrada

Out[∘]=
$$\frac{\sqrt{2-\sqrt{2}}}{2\sqrt{\frac{k}{m}}\ m}$$

In[∘]:= D[Sqrt[(2 - Sqrt[2]) * k / m], m]
⌊·· ⌊raíz cuad··· ⌊raíz cuadrada

Out[∘]=
$$-\frac{\sqrt{2-\sqrt{2}}\ k}{2\sqrt{\frac{k}{m}}\ m^2}$$

In[∘]:= errw2 = Sqrt$\left[\left(\frac{\sqrt{2-\sqrt{2}}}{2\sqrt{\frac{k}{m}}\ m} * ek\right)\text{^}2 + \left(\frac{\sqrt{2-\sqrt{2}}\ k}{2\sqrt{\frac{k}{m}}\ m^2} * em\right)\text{^}2\right]$
⌊raíz cuadrada

Out[∘]=
0.00616607

In[∘]:= (*w3*)
D[Sqrt[(2 + Sqrt[2]) * k / m], k]
⌊·· ⌊raíz cuad··· ⌊raíz cuadrada

Out[∘]=
$$\frac{\sqrt{2+\sqrt{2}}}{2\sqrt{\frac{k}{m}}\ m}$$

In[∘]:= D[Sqrt[(2 + Sqrt[2]) * k / m], m]
⌊·· ⌊raíz cuad··· ⌊raíz cuadrada

Out[∘]=
$$-\frac{\sqrt{2+\sqrt{2}}\ k}{2\sqrt{\frac{k}{m}}\ m^2}$$

In[∘]:= errw3 = Sqrt$\left[\left(\frac{\sqrt{2+\sqrt{2}}}{2\sqrt{\frac{k}{m}}\ m} * ek\right)\text{^}2 + \left(-\frac{\sqrt{2+\sqrt{2}}\ k}{2\sqrt{\frac{k}{m}}\ m^2} * em\right)\text{^}2\right]$
⌊raíz cuadrada

Out[∘]=
0.0148862

In[ ]:=
```
(*2*)
Clear[k, m, ek, em]
borra
k = 10.15003805;
ek = 0.000339653;
m = 0.25075;
em = 0.00065;
(*simetrico*)
D[Sqrt[k / m], k]
raíz cuadrada
```

General: 10.15003805` is not a valid variable.

Out[ ]=

$\partial_{10.15} 6.36229$

In[ ]:=
```
D[Sqrt[k / m], m]
raíz cuadrada
```

Out[ ]=

$$-\frac{k}{2 \sqrt{\frac{k}{m}}\, m^2}$$

In[ ]:= errw1 = Sqrt$\left[\left(\frac{1}{2 \sqrt{\frac{k}{m}}\, m} * ek\right)\text{^}2 + \left(\frac{k}{2 \sqrt{\frac{k}{m}}\, m^2} * em\right)\text{^}2\right]$

raíz cuadrada

Out[ ]=

0.00824692

```
(*aantismetrtico*)
```

In[ ]:=
```
D[Sqrt[3 * k / m], k]
raíz cuadrada
```

Out[ ]=

$$\frac{\sqrt{3}}{2 \sqrt{\frac{k}{m}}\, m}$$

In[ ]:=
```
D[Sqrt[3 * k / m], m]
raíz cuadrada
```

Out[ ]=

$$-\frac{\sqrt{3}\, k}{2 \sqrt{\frac{k}{m}}\, m^2}$$

In[ ]:= $errw2 = \text{Sqrt}\left[\left(\frac{\sqrt{3}}{2\sqrt{\frac{k}{m}}\ m} * ek\right)^{\wedge}2 + \left(\frac{\sqrt{3}\ k}{2\sqrt{\frac{k}{m}}\ m^2} * em\right)^{\wedge}2\right]$

raíz cuadrada

Out[ ]=

0.0142841

In[ ]:= em

Out[ ]=

0.00065